Finance Pulse: Integrated Banking Analytics & Risk Assessment Dashboard

A Case Study on Data-Driven Risk Management in Banking

Executive Summary

This case study presents a comprehensive analysis of a banking risk analytics project designed to minimize financial risk in lending operations. Through a systematic approach involving data processing, analysis, and visualization, the project delivers actionable insights that enable banks to make informed lending decisions based on applicant profiles. The solution employs a multi-stage data pipeline using MySQL for data processing, Python for exploratory data analysis, and Power BI for creating interactive dashboards that highlight key banking metrics across different customer segments.

1 Business Problem

Financial institutions face significant challenges in assessing lending risk, leading to potential revenue losses from loan defaults. The core business problem addressed in this project is:

> *How can banks use data analytics to minimize the risk of losing money while lending to customers?*

The project aims to solve this problem by developing a comprehensive understanding of risk factors and creating visualization tools that support decision-making processes for loan approvals.

2 Project Objectives

1. Analyze banking and customer data to identify risk patterns
2. Develop metrics and KPIs that assess lending risk across different customer segments
3. Create interactive dashboards to visualize financial data and support decision-making
4. Provide insights that help determine whether applicants are likely to repay loans

3 Methodology

3.1 Data Source and Structure

The project utilized a comprehensive banking dataset comprising multiple interrelated tables:

- Banking Relationship
- Client-Banking
- Gender
- Investment Advisor
- Period

These tables were connected through primary and foreign keys, creating a relational database structure that allowed for complex analyses across different dimensions of the banking business.

3.2 Technical Approach

The project followed a structured data pipeline:

3.2.1 Data Collection and Integration

- Sourced data from existing banking systems
- Imported into MySQL server for centralized management

3.2.2 Data Cleaning and Transformation

- Created derived columns for enhanced analysis:
  - *Engagement Timeframe*: Timeline of client-bank relationships
  - *Engagement Days*: Duration of client relationships calculated from joining date
  - *Income Band*: Categorization of clients by income level (<\$100K = Low, <\$300K = Mid, \$300K = High)
  - *Processing Fees*: Fee calculations based on fee structure (e.g., high fee = 0.05)

3.2.3 Exploratory Data Analysis with Python

- Analyzed patterns and correlations in the data
- Identified key risk factors and client segments

3.2.4 Dashboard Development with Power BI

- Implemented calculated measures using DAX formulas:
  - SUM for aggregations
  - DISTINCTCOUNT for unique value counts
  - SUMX for row-level calculations
  - SWITCH for conditional logic
  - DATEDIFF for time-based calculations

3.2.5 Visualization and Insight Generation

- Created four specialized dashboard views:
  - Home (overview)
  - Loan Analysis
  - Deposit Analysis
  - Summary Dashboard

4 Key Performance Indicators (KPIs)

The dashboard tracked several critical KPIs to measure banking performance and risk:

4.1 Client Metrics

- **Total Clients**: 1,499 distinct banking clients (Home Dashboard view)
- **Total Clients by Filter**: 676 clients (when filtered to specific banking relationships)
- **Engagement Account**: $3.53M representing client engagement value

4.2 Financial Metrics

- **Total Loan**: $2.19 billion (Home Dashboard) / $1.01 billion (filtered view)
- **Total Deposit**: $1.9 billion (Home Dashboard) / $869.93 million (filtered view)
- **Total Fees**: $78.71 million (Home Dashboard) / $36.71 million (filtered view)

4.3 Detailed Financial Metrics

- **Bank Loan**: $407.98 million in direct bank loans
- **Business Lending**: $599.58 million in loans to businesses
- **Bank Deposit**: $468.23 million in direct bank deposits
- **Savings Account**: $345.68 million (Home Dashboard) / $160.86 million (filtered view) in savings accounts
- **Foreign Currency Account**: $20.59 million in foreign currency holdings
- **Checking Account Amount**: $220.25 million in transactional accounts
- **Credit Cards Balance**: Part of Total Loan calculation
- **Total CC Amount**: $2.22K (Home Dashboard) / $1K (filtered view)

5 Key Findings and Insights

5.1 Income Band Analysis

- **Mid-Income Band** ($100K-$300K) clients represented the largest loan segment at $207.58M
- **Low-Income Band** (<$100K) clients had $109.5M in loans
- **High-Income Band** ($300K) clients showed the lowest loan amounts at $90.9M

This counter-intuitive finding suggests that middle-income clients may be the most significant market for banking loans, while high-income clients potentially rely less on traditional bank financing.

5.2 Geographical Distribution

- **European Clients** showed the highest loan ($181.51M) and deposit values ($0.70B)
- **Asian Clients** were the second-largest segment with $100.93M in loans and $0.39B in deposits
- **American Clients** had $68.5M in loans and $0.21B in deposits
- **African Clients** had $20.57M in loans and $0.08B in deposits
- **Australian Clients** had $36.46M in loans

This geographical distribution highlights important regional differences in banking behavior and potential market opportunities.

5.3 Account Type Distribution

- **Bank Deposits** ($468.23M) formed a significant portion of total deposits
- **Checking Accounts** ($220.25M) represented a substantial portion of client funds
- **Savings Accounts** ($160.86M) represented another significant deposit category
- **Foreign Currency Accounts** ($20.59M) showed lower but potentially strategic importance

5.4 Banking Type Insights

Private banks demonstrated a higher client count compared to other banking types, suggesting potential competitive advantages in client acquisition strategies.

6 Business Impact

The Banking Risk Analytics Dashboard delivers several significant business benefits:

1. **Enhanced Risk Assessment**: The dashboard enables more accurate evaluation of loan applicants based on multiple factors including income band, nationality, and banking behavior.
2. **Strategic Market Insights**: By revealing which customer segments represent the highest loan and deposit volumes, banks can tailor their marketing and product strategies to target high-value segments.
3. **Competitive Analysis**: Insights into banking types with higher client counts provide strategic direction for institutions looking to increase market share.
4. **Decision Support**: The comprehensive visualization of financial metrics provides decision-makers with at-a-glance insights that support both strategic planning and day-to-day operational decisions.
5. **Financial Performance Tracking**: The dashboard provides a centralized view of key financial metrics, allowing for ongoing monitoring of institutional performance.

7 Future Development Opportunities

Based on the project outcomes, several opportunities for future enhancement have been identified:

1. **Predictive Risk Modeling**: Develop machine learning models that predict loan default risk based on historical client data.
2. **Client Segmentation Refinement**: Further segment customers based on behavioral patterns to develop more targeted banking products.
3. **Real-time Data Integration**: Enhance the dashboard with real-time data feeds to provide up-to-the-minute insights for decision-making.
4. **Cross-selling Opportunity Analysis**: Analyze relationships between different banking products to identify cross-selling opportunities.
5. **Comparative Benchmarking**: Incorporate industry benchmarks to provide context for performance metrics.

8 Technical Implementation

The project utilized a modern data analytics stack:

- **Data Storage and Manipulation**: MySQL Server for relational data management
- **Data Analysis**: Python with libraries such as Pandas, NumPy, and Matplotlib for exploratory analysis
- **Data Visualization**: Power BI for interactive dashboard creation
- **DAX Formulas**: Custom calculations using Data Analysis Expressions
- **Documentation**: Comprehensive technical and business documentation

9 Conclusion

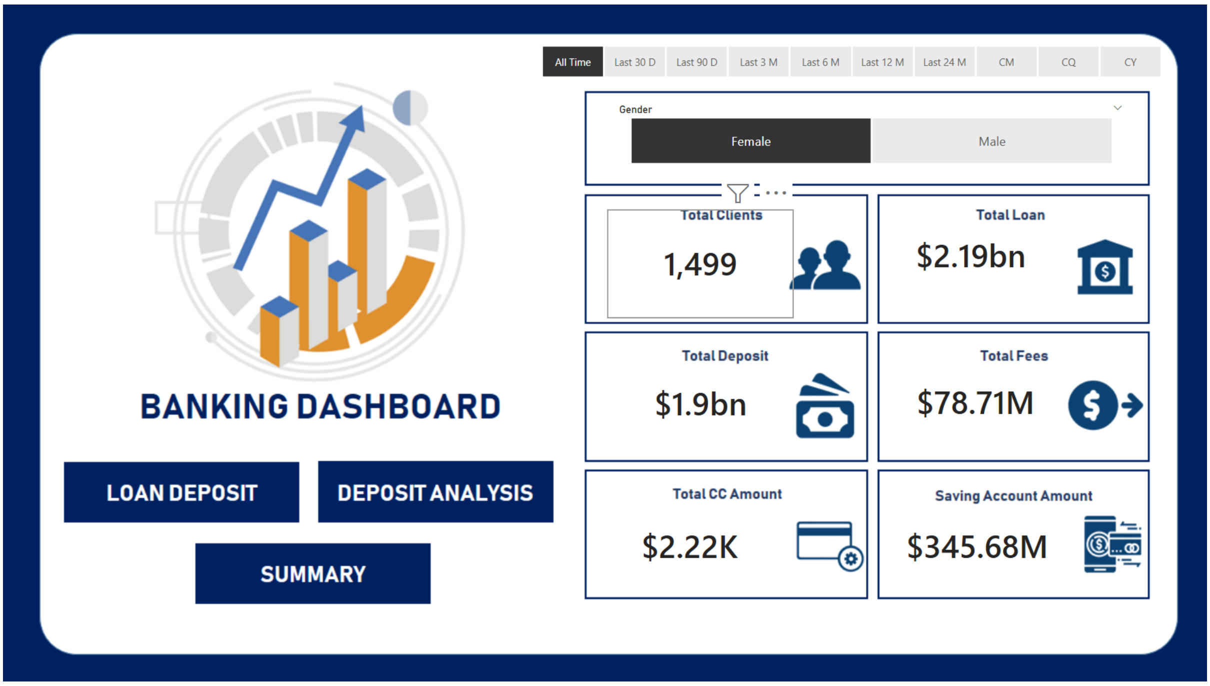
The Finance Pulse project demonstrates the significant value that data-driven analytics brings to financial risk management. By transforming raw banking data into actionable insights, the project enables more informed lending decisions that minimize financial risk while optimizing business opportunities.

The analysis revealed important patterns in banking behavior across different customer segments, particularly highlighting the significance of mid-income clients and regional variations in banking engagement. These insights provide a foundation for more targeted marketing strategies and product offerings.

By integrating data processing, analysis, and visualization into a cohesive solution, the project delivers a powerful tool for banking professionals that enhances decision-making capabilities and supports strategic business goals.

10 Appendix: Dashboard Screenshots and Analysis

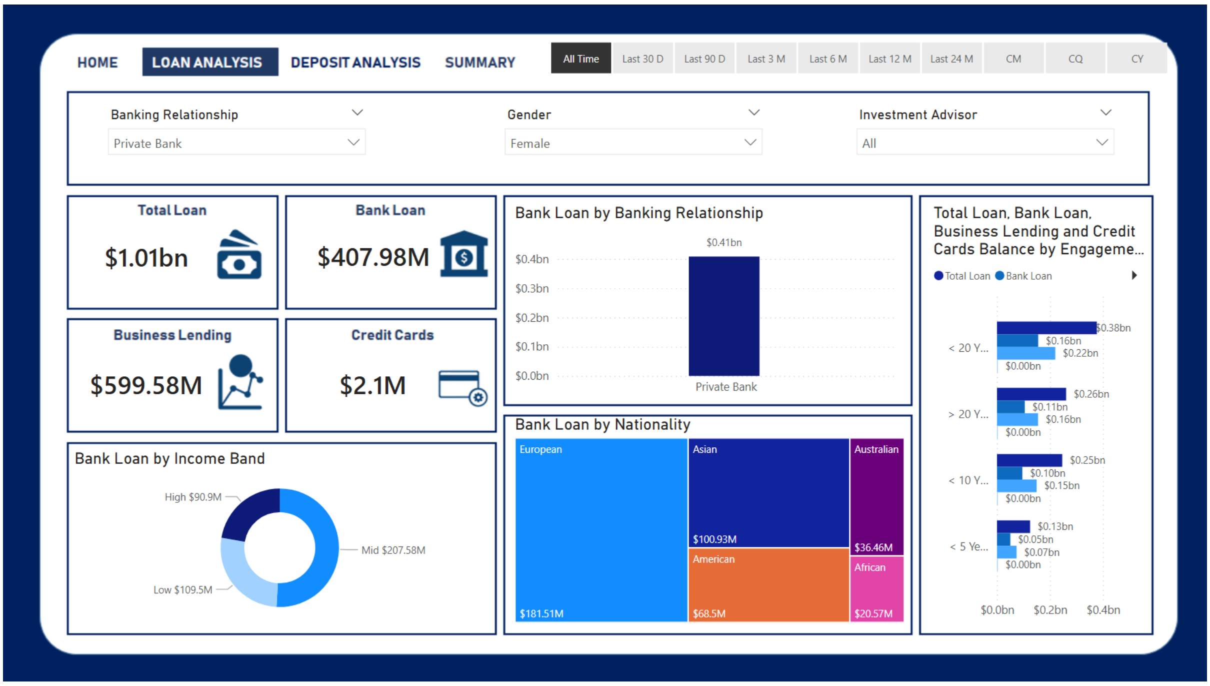
10.1 Home Dashboard



The Home Dashboard provides a comprehensive overview of all key banking metrics with a clean, professional interface featuring:

- Time period filters (All Time, Last 30D, Last 90D, etc.)
- Gender filter (showing data for Female clients)
- Key KPIs including Total Clients (1,499), Total Loan ($2.19bn), Total Deposit ($1.9bn), and Total Fees ($78.71M)
- Additional metrics for Total CC Amount ($2.22K) and Saving Account Amount ($345.68M)
- Navigation menu to access Loan Deposit, Deposit Analysis, and Summary sections

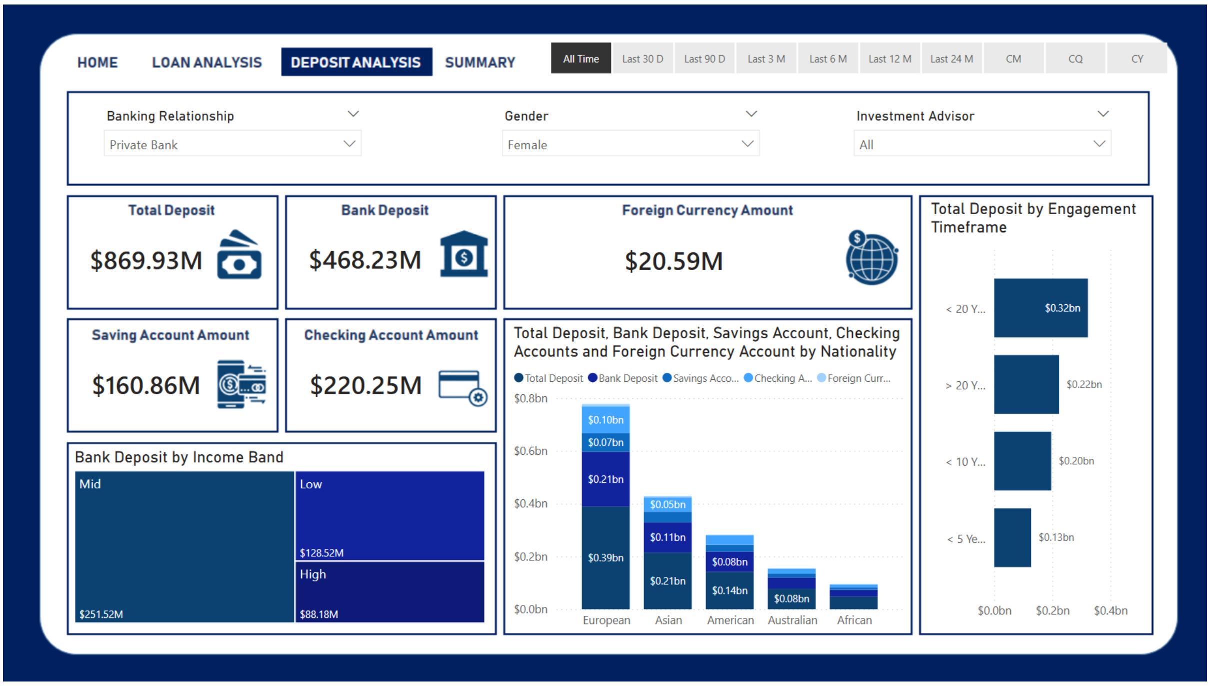
10.2 Loan Analysis Dashboard



The Loan Analysis page offers detailed insights into lending patterns with:

- Multiple filters for Banking Relationship, Gender, and Investment Advisor
- Core metrics showing Total Loan ($1.01bn), Bank Loan ($407.98M), Business Lending ($599.58M), and Credit Cards ($2.1M)
- Bank Loan by Income Band visualization (donut chart) showing the distribution across High ($90.9M), Mid ($207.58M), and Low ($109.5M) income segments
- Bank Loan by Nationality treemap visualization showing European clients ($181.51M) with the highest loans, followed by Asian ($100.93M), American ($68.5M), Australian ($36.46M), and African ($20.57M) clients
- Loan metrics by Engagement Timeframe visualization showing how loan distribution varies by client relationship duration

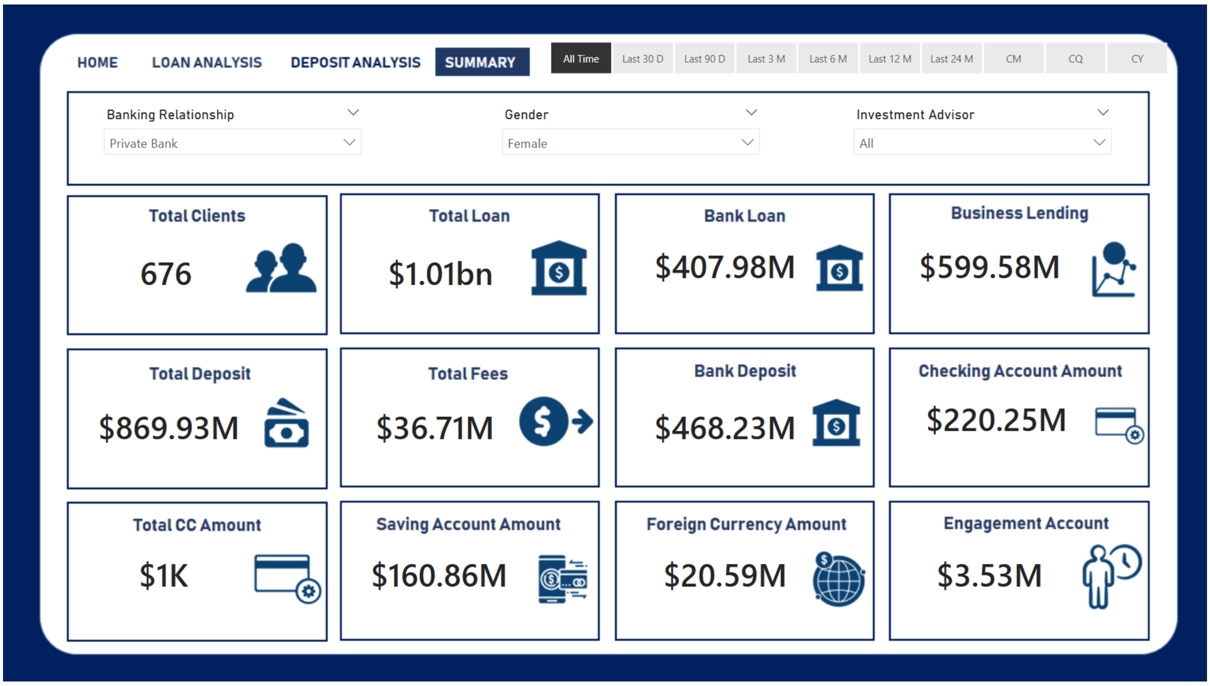
10.3 Deposit Analysis Dashboard



The Deposit Analysis page focuses on depositor behavior with:

- Complete deposit metrics including Total Deposit ($869.93M), Bank Deposit ($468.23M), Foreign Currency Amount ($20.59M)
- Savings Account Amount ($160.86M) and Checking Account Amount ($220.25M)
- Bank Deposit by Income Band visualization showing Mid-income clients with highest deposits ($251.52M) followed by Low ($128.52M) and High ($88.18M)
- Comprehensive stacked bar chart showing deposit distribution by nationality with European clients leading in all deposit categories
- Total Deposit by Engagement Timeframe showing relationship between client tenure and deposit amounts

10.4 Summary Dashboard



The Summary Dashboard consolidates all critical metrics in a single view:

- Total Clients (676 for the selected filter)
- Complete financial metrics: Total Loan ($1.01bn), Bank Loan ($407.98M), Business Lending ($599.58M)
- Deposit indicators: Total Deposit ($869.93M), Bank Deposit ($468.23M), Checking Account Amount ($220.25M)
- Additional metrics: Total Fees ($36.71M), Total CC Amount ($1K), Saving Account Amount ($160.86M), Foreign Currency Amount ($20.59M), and Engagement Account ($3.53M)
- Clean, icon-based design for quick visual identification of metric categories